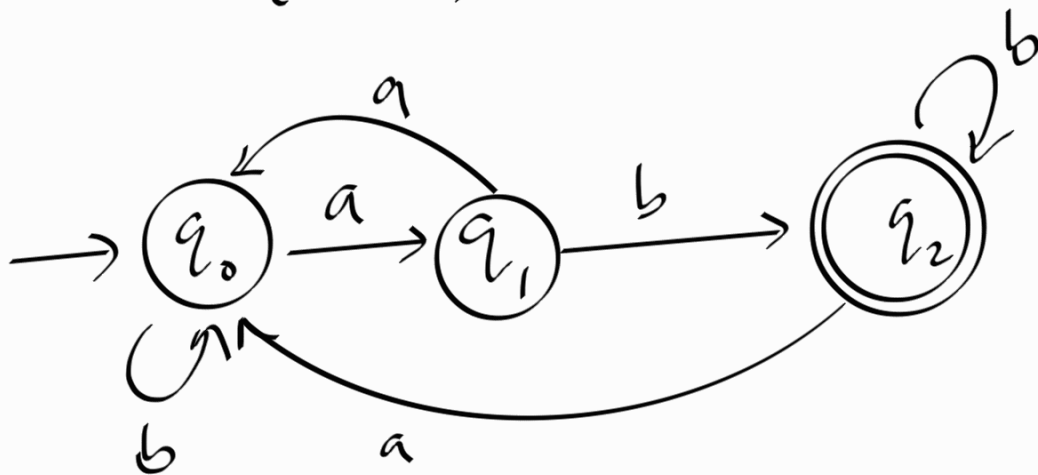


Let's do some examples

1. Design a DFA that accepts the following language.

$$C = \{ w \mid w \text{ has an odd \# of } a\text{'s and ends with } b \}$$

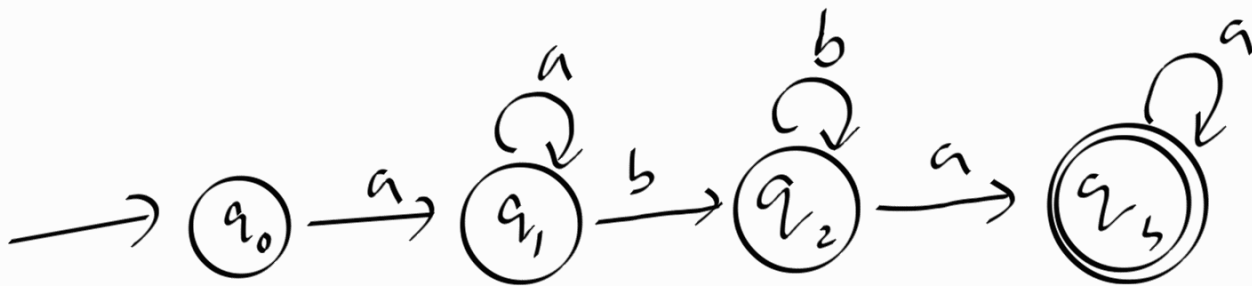
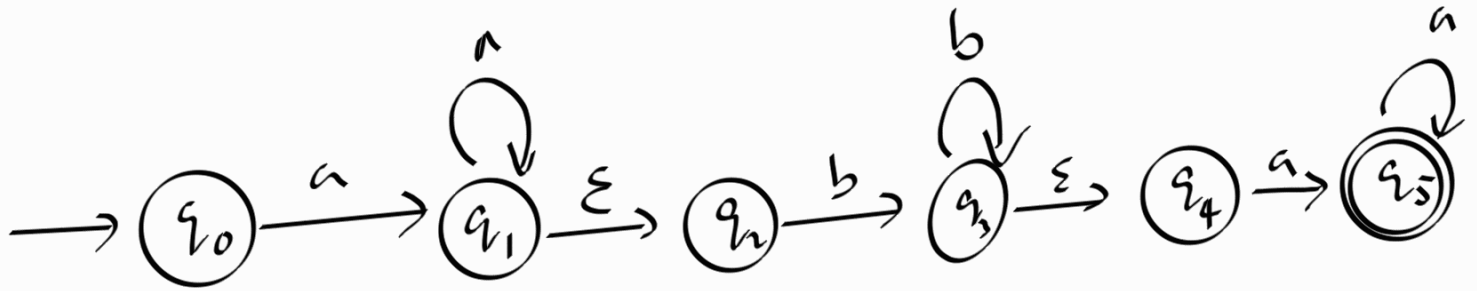
$$\Sigma = \{ a, b \}$$



Ex2: Design an NFA for

$$D = a^+ b^+ a^+$$

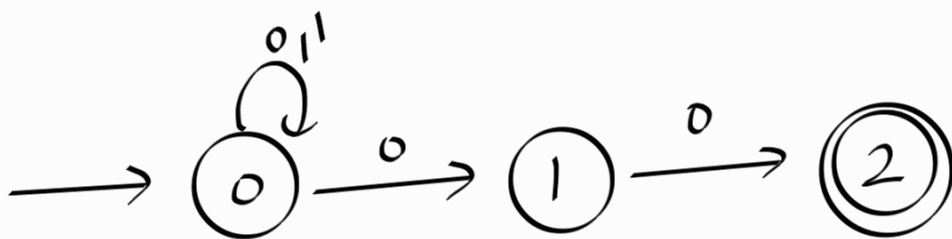
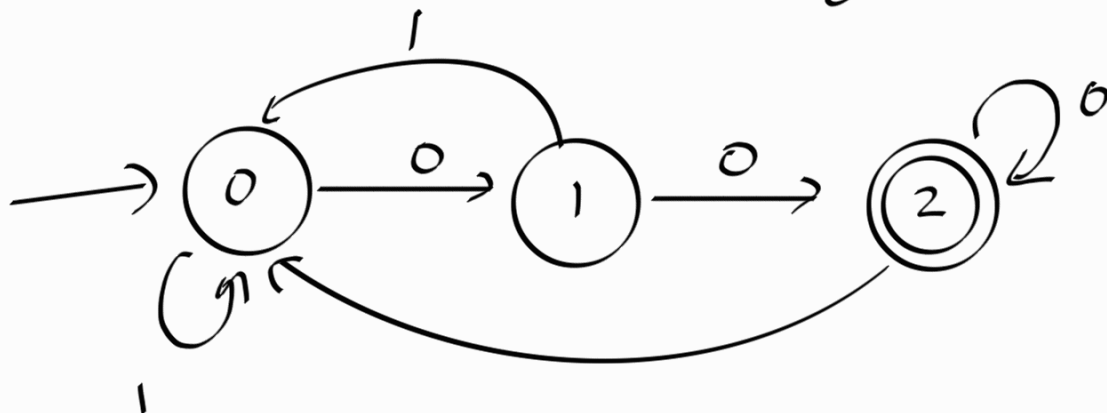
$$\Sigma = \{a, b\}$$



Ex 5: Design an NFA for

$$E = \{ w \mid w \text{ ends with } 00 \}$$

$$\Sigma = \{0, 1\}$$



00100

Definition

Two machines are equivalent if they accept the same language.

Theorem 1.39

Every NFA has an equivalent DFA.

Proof: Let $N = (Q, \Sigma, \delta, q_0, F)$ be an NFA that recognize language A .

We'll construct a DFA M , $M = (Q', \Sigma, \delta', q'_0, F')$

a) No ϵ -transitions in N .

1. $Q' = P(Q)$

2. $\Sigma' = \Sigma$

3. $\forall R \in Q', a \in \Sigma, \delta'(R, a) = \bigcup_{r \in R} \delta(r, a)$

4. $q'_0 = \{q_0\}$

5. $F' = \{R \in Q' \mid R \text{ contains an accept state in } N\}$

The basic idea for using the powerset is to simulate every possible execution path an NFA can take.

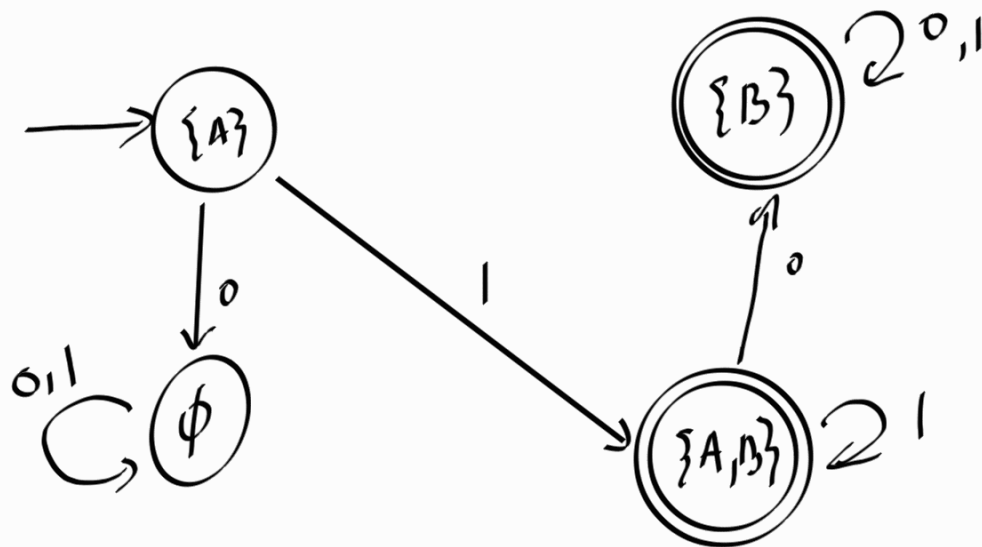
Let's look at an example

NFA N :



An example
w.o
 ϵ -transitions

DFA $M : Q' = P(Q), Q' = \{\emptyset, \{A\}, \{B\}, \{A, B\}\}$



$$\delta'(\{A\}, 0) = \bigcup_{r \in \{A\}} \delta(r, 0) = \emptyset$$

$$\delta'(\{A\}, 1) = \bigcup_{r \in \{A\}} \delta(r, 1) = \{A, B\}$$

$$\delta'(\{B\}, 0) = \bigcup_{r \in \{B\}} \delta(r, 0) = \{B\}$$

$$\delta'(\{B\}, 1) = \bigcup_{r \in \{B\}} \delta(r, 1) = \{B\}$$

$$\begin{aligned}\delta'(\{A, B\}, 0) &= \bigcup_{r \in \{A, B\}} \delta(r, 0) = \delta(A, 0) \cup \delta(B, 0) \\ &= \emptyset \cup \{B\} \\ &= \{B\}\end{aligned}$$

$$\begin{aligned}\delta'(\{A, B\}, 1) &= \bigcup_{r \in \{A, B\}} \delta(r, 1) = \delta(A, 1) \cup \delta(B, 1) \\ &= \{A, B\} \cup \{B\} \\ &= \{A, B\}\end{aligned}$$

$$\delta(\emptyset, 0) = \bigcup_{r \in \emptyset} \delta(r, 0) = \emptyset$$

$$\delta(\emptyset, 1) = \bigcup_{r \in \emptyset} \delta(r, 1) = \emptyset$$